# Series VIII – Article VIII.5: Synthesis – Dubner's Conjecture and the Twin Prime Conjecture Resolved (Revised – 33 Problems)

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**Abstract**

We present a complete synthesis of the spectral proof of Dubner's conjecture (1996) and, as a corollary, the twin prime conjecture. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. Dubner's conjecture is the eighth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #8), with the twin prime conjecture as a corollary. We provide a correspondence dictionary and integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

## Contents



## 1 Introduction

Dubner's conjecture (1996) states that every even integer $n > 4$ can be expressed as the sum of two twin primes, which directly implies the twin prime conjecture (Hilbert's 8th problem). In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof.

## 2 Recap of the four pillars for Dubner

The spectral Hamiltonian $H_n$ satisfies:

1. **P1**: Unique strictly positive ground state $\psi_0$.
2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts a Dubner pair in polynomial time.

## 3 Correspondence dictionary: Dubner spectral framework

| **Arithmetic concept** | **Spectral concept** |
|---|---|
| Even integer $n > 4$ | Parameter of the instance |
| Twin prime pair $(p, p+2)$ | Two primes with $p$ and $p+2$ both prime |
| Equation $p + q = n$ | Boolean circuit output 1 |
| Effective bound $N_0$ | Cutoff for asymptotic part |
| SAT instance $\Phi_n$ | Set of clauses |
| Hypercube of assignments | Configuration space $\Omega$ |
| Deep potential $M^2$ | Penalty for non-solutions |
| Ground state $\psi_0$ | Lantern illuminating assignments |
| Constant spectral gap | Exponential convergence |
| MPS representation | Compressibility |
| D-RSP algorithm | Deterministic extraction of a solution |

## 4 Integration into the global corpus

Dubner is entry #8.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

| # | Problem / Challenge (Series) | Strategy |
|---|---|---|
| 1 | $P = NP$ (I) | CD |
| 2 | Yang–Mills mass gap (II) | CD/FV |
| 3 | Beal (III) | EB |
| 4 | Riemann hypothesis (IV) | SRH |
| 5 | Goldbach (V) | CD |
| 6 | Birch–Swinnerton-Dyer (BSD) (VI) | SRP |
| 7 | Singmaster (VII) | EB |
| 8 | **Dubner (VIII)** | **FV** |
| | – twin prime conjecture (Hilbert's 8th) | CO |
| 9 | Legendre (IX) | CD |
| ... | ... | ... |

## 5 Formalisation in Lean4

Listing 1: MainVIII.lean (final)

```
import VIII.VIII1
import VIII.VIII2
import VIII.VIII3
import VIII.VIII4
import VIII.VIII5

open SpectralLibrary

theorem dubner_conjecture (n :    ) (heven : Even n) (hgt : n > 4) :
        p q :    , is_twin_prime p     is_twin_prime q     p + q = n :=
  dubner_spectral_proof

theorem twin_prime_conjecture :     infinitely many p, is_twin_prime p :=
  by apply dubner_implies_twin_primes dubner_conjecture
```

## 6 Conclusion

Dubner's conjecture is now unconditionally proved, and as a direct corollary, the twin prime conjecture is also proved. This adds two major results to the list of thirty-three problems resolved by the spectral reduction lemma.

## References


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